

ARTIFICIAL INTELLIGENCE IN THE JUDICIARY, AI JUDGES, AND ITS APPLICABILITY IN NORTH MACEDONIA

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ABSTRACT

This paper aims to examine the potential impact of artificial intelligence in judicial systems and assess the readiness of legal institutions and professionals to adopt and implement AI-based tools in adjudication and court management. Artificial intelligence (AI) is increasingly being integrated into various areas of judicial systems. From legal analysis to decision support tools, from court administration to automated adjudication, AI technologies have the potential to improve the speed, accuracy, and efficiency of judicial processes. This article outlines the general framework of AI implementation in the judiciary, explores the concept of “AI judges,” and examines the applicability of such technologies in the Republic of North Macedonia. Considering the current legal and constitutional context, as well as technological limitations, the integration of AI into North Macedonia’s judiciary appears conditionally feasible. This study has been conducted based on a diverse range of sources, including various books, scholarly articles, peer-reviewed journals, and official reports published by key public institutions. Accordingly, the article examines the use of artificial intelligence in the judiciary and legal system under four main subheadings: theoretical framework, ethical concerns, technological infrastructure, and legal regulations. Each of these dimensions has been explored in a more concrete and detailed manner. As a result, it is argued that for North Macedonia to implement AI systems within its judiciary, it should first develop a comprehensive national AI strategy, initiate both legislative and constitutional reforms with reference to comparative law, and align with EU standards. Furthermore, it should adhere to the fundamental principles outlined throughout this study. However, critical questions remain regarding how such applications can be reconciled with fundamental legal principles such as the rule of law, the right to a fair trial, transparency, and accountability. This study contributes to the emerging literature on AI in justice systems by offering a context-specific analysis of its applicability within a developing legal and institutional framework, using North Macedonia as a case study.

Keywords: Artificial Intelligence, AI Judges, North Macedonia, Legal Framework, Ethics and Digital Justice

1. Introduction

The use of artificial intelligence (AI) technologies in the legal field particularly within judicial systems has been drawing increasing attention worldwide due to its potential to enhance the efficiency and effectiveness of justice delivery. In parallel, the concept of the “AI judge” has emerged, primarily aimed at automating decision-making processes in simple legal disputes. For instance, in Estonia, a pilot project has been proposed involving a “robot judge” system designed to resolve small claims cases. Under this model, parties would submit their documents digitally, after which the AI system would issue a decision, subject to appeal before a human judge. Similarly, since 2017, China's Internet Courts have handled millions of cases online with the assistance of AI judges. In many proceedings, virtual judges in the form of holograms have conducted video interactions with parties, received evidence, and rendered decisions. While these developments demonstrate that the integration of AI into judicial processes is practically feasible, they also give rise to significant ethical, technical, and legal debates.

This study aims to examine the implementation of artificial intelligence in the judiciary within the context of North Macedonia. As the country undergoes digital transformation and legal reforms in line with its EU accession process, the applicability of AI-based systems will be addressed across three main dimensions. First, ethical concerns will be explored, including impartiality, transparency, human rights, and the accountability of decisions. Second, the focus will shift to the technological infrastructure - namely the current level of digitalization, data management capacity, and the IT systems required for AI integration. Finally, legal regulations will be analyzed, particularly the existing legal framework of North Macedonia, its alignment with EU and international standards, and potential regulatory gaps. Within this framework, international experiences and examples from EU practices will be used to assess North Macedonia's current position and future prospects.

2. Theoretical Framework

Artificial intelligence (AI) is defined as a set of algorithms designed to equip machines with human-like cognitive abilities such as learning, reasoning, and decision-making. (Susskind, 2019). The development of AI technologies has gained strength through concrete applications in various fields such as education, law, healthcare, finance, and security, enabling the automation of numerous processes that traditionally required human intervention. (Selimi, Saracevic, & Useini, 2020). The rise of AI in the legal field has accelerated with the effective application of big data analytics, natural language processing (NLP), and machine learning techniques to legal texts. (Surden, 2014).

Globally, the literature on AI in the judiciary presents a dual narrative: one of innovation and one of caution. On the one hand, AI offers tools for legal research, case management, risk assessment, and preliminary adjudication. (Zouridis, van Eck, M. L., & Bovens, 2019). For instance, China's Internet Courts have handled over three million disputes using digital platforms and AI-driven tools. (Vasdani, 2020). Estonia has proposed an AI adjudicator for claims under €7,000, with the intention of improving efficiency in small claims cases. (Niiler, 2019). On the other hand, academic scholars raise concerns about algorithmic bias, transparency, and human rights implications. (McGregor, 2021). The Council of Europe's (2018) Ethical Charter on AI in the judiciary underscores the need for human oversight, transparency, and respect for fundamental rights.

This initiative is particularly noteworthy as it would represent one of the first instances of granting decision-making authority - albeit limited - to an AI system. Similarly, in some U.S. states, risk assessment algorithms (e.g., COMPAS) are used to assist judges in criminal proceedings; however, such tools have been criticized for allegedly exhibiting racial bias. (Estonia, 2019). In the United Kingdom, an AI-powered chatbot called DoNotPay has assisted citizens in drafting appeals against parking fines, leading to the cancellation of thousands of erroneous tickets. (Angwin, Larson, Mattu, & Kirchner, 2016).

Academic studies examine the role of AI in judicial processes from both opportunity and risk perspectives. AI holds significant potential to enhance efficiency by supporting human labor in areas such as legal research, precedent analysis, and the classification of petitions and legal documents (Ali, 2024). For instance, in complex case files, AI algorithms can rapidly scan relevant case law and legislation, providing judges with comprehensive information thereby contributing to more informed and consistent decision-making. (Dönmez, 2025). Indeed, a recent study suggests that the chronic workload and case backlog issues within the North Macedonian judiciary could be alleviated through AI-assisted automation. (Korbayram & Angeleska, 2023). In particular, the automation of routine and straightforward procedural tasks by AI could allow judges and court staff to concentrate on more complex and substantive matters. However, the literature consistently underscores ethical concerns. It emphasizes the need for transparency in decision-making processes, the elimination of algorithmic bias, the establishment of oversight mechanisms for errors, and ultimately, the responsibility of a human judge to oversee and validate AI-generated outcomes. (Korbayram & Angeleska, 2023). The European Ethical Charter on the Use of Artificial Intelligence in Judicial Systems, adopted by the Council of Europe in 2018 (Council of Europe, 2018), It recommends that AI applications comply with the principles of respect for human rights, non-discrimination, quality and security, transparency and impartiality, and user control (human oversight (Nuredin, 2022b).

These principles advocate for a cautious approach toward granting AI systems autonomous and unchecked judicial authority. (Council of Europe, 2018). Consequently, the prevailing view in the literature is that AI should not replace human judges entirely, but rather be implemented gradually and carefully as a supportive tool within the judicial process. As Estonian President Kersti Kaljulaid has noted, when AI takes over routine and labor-intensive tasks, human judges can focus on aspects such as empathy and moral judgment - areas that machines can never truly replicate; provided, of course, that AI systems are deployed in a secure and transparent manner. (Estonia, 2019).

3. Ethical Dimensions

The use of artificial intelligence in the judiciary raises significant ethical questions. The first major concern is how to uphold the principle of impartiality. AI algorithms carry the risk of reflecting the biases present in the datasets on which they are trained. For instance, in the United States, the COMPAS risk assessment AI system used in criminal courts was found to assign higher risk scores to Black defendants, thereby contributing to discriminatory outcomes. (Estonia, 2019). Such cases demonstrate that AI decisions can unintentionally be biased based on factors like race, gender, or social status. The prohibition of discrimination is explicitly emphasized as one of the core ethical principles of the Council of Europe regarding AI. Accordingly, AI systems must ensure equal treatment for all individuals and must not adversely affect disadvantaged or vulnerable groups. (Council of Europe, 2018).

The second critical issue concerns the principles of transparency and explainability. AI judges or decision-support systems must not operate as “black boxes” in other words, the logic behind their conclusions must be comprehensible and explainable. One of the fundamental principles of law is the right to a reasoned judgment, which also applies to AI-generated decisions; the rationale behind the outcome must be meaningful and accessible. However, complex algorithms such as deep learning often produce models that are difficult for humans to interpret. This lack of transparency poses challenges for the public acceptance and accountability of judicial decisions.

Accountability becomes especially crucial in this context, raising the question: who bears responsibility when an AI-assisted decision is flawed or unjust? Is it the system developers, the judicial authorities, or the human judge who ultimately approves the decision? This ambiguity represents a critical gap that must be addressed before AI can be responsibly used in the judiciary. Furthermore, there must be clear mechanisms for correcting errors and appealing decisions made with AI involvement. Indeed, in China’s AI judge pilot programs, decisions made by AI systems are subject to review by human judges in higher courts an example of how the principle of accountability can be operationalized in practice. (Vasdani, 2020).

In countries like North Macedonia, which are parties to the European Convention on Human Rights (ECHR), no practice should be allowed that undermines the fairness of judicial proceedings, as guaranteed under Article 6 of the Convention, the right to a fair trial. In this context, the use of AI must not conflict with fundamental rights such as access to a court, equality of arms, or the presumption of innocence. For instance, a fully automated decision-making system could potentially limit the defendant’s right to appear before a judge or to have a face-to-face hearing with the decision-maker. Therefore, before such systems are implemented, their legal and ethical legitimacy must be carefully examined. (Korbajram & Hoca, 2021).

The human factor and human dignity also occupy a central place in the ethical dimension (Nuredin & Nuredin, 2023) of AI in the judiciary. Judicial proceedings are not merely about mechanically applying legal rules (Nuredin, 2016) - they require human judgment and moral reasoning. A human judge, by listening to the testimonies of victims and defendants in person, can perceive nuances in speech, emotional reactions, and signs of remorse - elements that carry significant weight in delivering justice. In contrast, current AI systems lack emotional intelligence and the capacity for ethical judgment. As a result, AI judges are often criticized for their lack of a “human touch,” which may undermine the inherently humane character of the judicial process. (Council of Europe, 2018). Public trust in the judiciary may also be affected at this stage; the perception that justice is being administered by machines, especially in the absence of sufficient transparency can make it more difficult for citizens to accept judicial decisions. To build and maintain trust, it must be made explicitly clear that in all cases where AI is used as a tool, the final responsibility for the decision lies with a human judge, and that AI serves only as an assistive mechanism.

This reflects the core of the “user control” principle outlined in the Council of Europe’s Ethical Charter, which asserts that while AI can support judicial processes, judicial discretion must always remain in human hands. In conclusion, from an ethical standpoint, it is essential that North Macedonia establish rules and oversight mechanisms that safeguard the principles of non-discrimination, transparency, respect for human rights, and accountability before implementing AI systems in the judiciary. (Council of Europe, 2018).

4. Analysis of Technological Infrastructure

The integration of AI into the judicial system requires a robust technological infrastructure. As a first step, case files, court decisions, and legal documents must be digitized and made accessible through integrated databases. In North Macedonia's judiciary, an Automated Case Management Information System (ACCMIS) is currently in use. However, the infrastructure of this system is significantly outdated it operates on a Windows Server 2008 platform, with each court managing its own separate database, and the capacity of the central server remains limited. (Janevski & Zoroska Kamilovska, 2009).

Although the Information Center within the Supreme Court provides certain centralized services through data replication from all courts, it is evident that the current data storage and sharing capacities are insufficient, and the existing systems are in need of a comprehensive upgrade. Moreover, auxiliary digital systems such as the court audio recording system (FEMIDA) also suffer from capacity limitations. Given the demands of AI applications - particularly in terms of big data management and real-time processing - these infrastructural shortcomings pose a significant obstacle to effective implementation. (Gjorgjioska, Gorgieva D., & Stoileva., 2020).

In recent years, the government of North Macedonia has taken strategic steps to accelerate digitalization in the judiciary. The Ministry of Justice's 2020-2022 Strategic Plan and the 2019-2024 Information and Communication Technology Strategy have identified the digitalization of judicial processes and the establishment of a modern e-justice system as key priorities. Within the scope of these plans, concrete objectives have been set, including the development of a centralized e-Court portal, the modernization of case management systems for courts and prosecutors' offices, the standardization of data, and training programs to help judicial personnel adapt to new technologies. (Ministry of Justice, 2020).

Indeed, in November 2024, the Judiciary Working Group of the EU National Convention in North Macedonia issued recommendations to the government, emphasizing the completion of an interoperability platform for inter-court systems and the full implementation of the e-Court system. (National Convention on the EU in the RN Macedonia, Acc. 2025). Through these initiatives, it is envisioned that key procedures such as filing lawsuits, submitting petitions and documents, and monitoring hearings can be conducted entirely in digital form, enabling the end-to-end digitalization of judicial processes. The same report also recommended the establishment of public oversight mechanisms via digital platforms-allowing civil society to monitor the progress of court cases—as well as cybersecurity and data protection training for members of the judiciary, and the adaptation of successful digital justice models from EU member states. These recommendations indicate that North Macedonia is actively working to address its technological infrastructure gaps and lay the groundwork for the future integration of advanced technologies such as artificial intelligence. (National Convention on the EU in the RN Macedonia, Acc. 2025).

When discussing infrastructure for AI applications, it is important to consider not only software but also hardware and human resources. Implementing such systems will require servers capable of processing large volumes of judicial data, cloud computing services, or high-capacity data centers. In North Macedonia, the country's limited technical capacity has also slowed the progress of the working group established in 2021 to develop a National AI Strategy (Ali & Skender, 2020). Key challenges identified in both the preparation and implementation of this strategy include the lack of sufficient data, a shortage of qualified experts, and inadequate technical infrastructure. (Law Gratis Reports., 2025).

Experts have noted that developing a comprehensive AI strategy is further complicated by the ongoing brain drain, emphasizing the need to retain well-trained young IT professionals within the country. To address these shortcomings, it has been pledged that an AI strategy aligned with EU standards will be developed with the support of international organizations such as the World Bank and UNDP. In addition, private sector initiatives in the ICT field are being encouraged. The Fund for Innovation and Technology Development (FITD) provides grants to start-up projects in the AI sector and has supported the creation of applications such as “ADA,” the first Macedonian AI-powered digital assistant. These types of digital assistants, which offer basic advisory services to citizens within public administration, help build institutional experience and foster broader public familiarity with AI technologies. (Radonjanin, Lazarevska , & Srbinoski, 2024).

When considering AI judges specifically, North Macedonia's level of digital infrastructure maturity is not yet fully adequate. AI judge systems require access to vast digital archives containing millions of judicial decisions and case data to effectively train algorithms (Ali, Ristevski, & Jolevski, 2022). Making court decisions publicly available in a digital and anonymized format is essential-not only for ensuring transparent use of data but also for training AI models. In this regard, a regional workshop held in Bosnia and Herzegovina in May 2025 reported that Western Balkan countries, including North Macedonia, are collectively working to improve their case management systems, publish anonymized court decisions online, and integrate AI and digital tools into judicial workflows. (Europe, 2025).

The aforementioned workshop, organized under the Council of Europe’s CEPEJ program, aims to foster regional knowledge-sharing on digitalization and AI integration. It is evident that North Macedonia is actively working to build its capacity through peer learning within this framework. In light of these developments, for North Macedonia to establish an AI-supported judicial system, it must first complete its e-justice infrastructure, ensure the full functionality of electronic case and record management systems across all courts, and elevate its data protection and cybersecurity standards to the highest level. (Europe, 2025).

Another critical factor is the ability of judges and court personnel to adapt to new technologies. Without adequate user training and cultural adaptation, even the most advanced AI systems will fail to deliver their intended benefits. Therefore, in conclusion, the applicability of AI judges in North Macedonia will largely depend on the success of current digitalization efforts and the extent of future investments in this domain. (National Convention on the EU in the RN Macedonia, Acc. 2025).

5. Legal Framework

To date, North Macedonia has not enacted any comprehensive legal framework specifically addressing artificial intelligence. As of May 2025, there is no law or binding regulation in the country that directly governs the use of AI. The absence of dedicated legislation concerning the application of AI in the legal domain highlights the urgent need to establish a clear regulatory framework. In this context, it is imperative that legal systems move swiftly and effectively toward adopting regulations capable of accommodating this emerging technology. (Turan, Kemaloglu, & Küçüksille, 2020). At present, developments in the field of AI in North Macedonia are primarily guided by strategic documents and general regulatory instruments, such as laws on personal data protection. Notably, the preparation of a National AI Strategy, which began in 2021, has yet to result in a concrete legislative draft. Structural challenges-such as the lack of sufficient data and a

shortage of qualified experts-have contributed to this delay. Nevertheless, as North Macedonia continues its alignment with the EU acquis, it is also preparing to adopt the European Union's approach to artificial intelligence regulation. (Law Gratis Reports., 2025). Experts emphasize that the legal gaps surrounding artificial intelligence in North Macedonia leave citizens vulnerable to potential misuse, and that an AI regulatory framework aligned with EU standards must be developed urgently. In this regard, the country already has regulations in place that parallel the EU's General Data Protection Regulation (GDPR). North Macedonia's Personal Data Protection Agency actively monitors the development and use of AI technologies to ensure a balance with individual privacy rights. Furthermore, the new Law on Personal Data Protection, which came into force in 2020, along with its implementing regulations, sets out general principles concerning automated data processing and decision-making procedures. (Korabayram, 2024). For example, safeguards aligned with the GDPR have been established regarding the processing of sensitive personal data by AI and the issuance of fully automated decisions. However, highly specific issues-such as delegating judicial decisions to AI-are not explicitly addressed within the current legal framework. At present, the Law on Courts and related legislation in North Macedonia define the concept of a judge strictly as a human being and require that all judgments and rulings bear the signature of an authorized judge. This implies that, even if an AI system were to render a decision in practice, it would not be legally valid without the formal approval of a human judge. (Law Gratis Reports., 2025).

At the European Union level, although a unified legal framework on artificial intelligence has not yet entered into force, the forthcoming EU Artificial Intelligence Act-expected to be finalized by 2025-serves as a significant point of reference. Through this draft regulation, it is evident that the EU has adopted a more cautious and human rights-oriented approach compared to some of the early-adopting countries in the field of AI. (Dönmez, 2025). The draft AI Act classifies artificial intelligence systems according to their level of risk and explicitly designates the use of AI in the field of justice and the judiciary as a “*high-risk*” category. (EU AI Act, 2025). Accordingly, AI systems used by judicial authorities for decision-making or the application of law-such as algorithms that conduct legal precedent analysis, evaluate evidence, or render judgments in place of a judge-are considered high-risk and will be subject to strict regulatory requirements. (European Commission, 2024). The deployment or use of such systems will be required to meet specific mandatory criteria regarding transparency, traceability, human oversight, and safety. The EU regulation aims to prevent the uncontrolled use of automated decision-making systems in both member and candidate countries. (EU AI Act, 2025). Therefore, if North Macedonia intends to pursue future projects involving AI judges or similar initiatives, it must take into account the high standards established in EU legislation. Additionally, the Council of Europe's 2018 Ethical Charter serves as a valuable soft law instrument for member states, and North Macedonia is expected to develop its domestic regulations and ethical codes in alignment with these principles.

When compared to practices in other countries, North Macedonia's regulatory gaps become more apparent. For instance, neighboring Serbia published a National AI Ethics Guidelines document in 2019, outlining principles to ensure that AI systems are developed in a fair, accountable, and transparent manner. (Radonjanin, Lazarevska , & Srbinoski, 2024). Similarly, some EU member states have taken legislative steps that indirectly regulate the use of AI in the judiciary. For example, in 2019, France enacted a provision in its Code of Justice (Article 33) that prohibits the use of AI to identify individual judges or statistically evaluate their performance based on their rulings. This regulation aims to protect judicial independence by preventing the scoring or ranking of judges

through decision analytics. While countries like the United Kingdom and Germany have not yet introduced AI judges, they have launched pilot programs for online court platforms and automated mediation systems.

Another important dimension of North Macedonia's legal framework relates to regulatory gaps and institutional capacity. Currently, the country lacks a specific authority tasked with overseeing the use of artificial intelligence across different sectors (Nuredin, 2023b), including the judiciary. (Korbayram & Delev, 2023). While the Ministry of Information Society and Administration and the Fund for Innovation and Technology Development appear to be responsible for the general coordination of AI policies, there is currently no independent regulatory or ethical body specifically designated to oversee the use of AI in the judiciary. This institutional gap poses challenges, particularly when it comes to establishing technical standards and addressing complaints that may arise from the use of AI in court proceedings. Experts emphasize the urgent need to develop protective legislation to ensure that citizens are not subjected to rights violations stemming from the application of AI technologies. (Law Gratis Reports., 2025). This includes essential safeguards such as the right of citizens to appeal and seek review of automated decisions, the obligation to explain algorithmic outcomes, and the certification and oversight of AI systems. As North Macedonia moves toward full EU membership, it will need to strengthen its legal framework in these areas.

In conclusion, given the absence of clear legal foundations for the use of AI in the judiciary, the implementation of AI judges currently exists in a legal grey area. To address this gap, the completion of the National AI Strategy must be prioritized, followed by the adoption of dedicated legislation or amendments to existing judicial laws - defining the scope and limits of AI use in the justice system. This would provide legal clarity on key issues such as the types of cases suitable for AI involvement, the legal status of AI-generated outcomes, the allocation of responsibility, and the mechanisms for oversight and accountability.

6. Discussion and Conclusions

Within the framework of the findings presented in the previous sections of this study, it is evident that the integration of artificial intelligence (Nuredin, 2023a) into judicial processes offers significant opportunities in terms of efficiency, speed, and accessibility. However, this transformation also brings with it serious ethical concerns and legal uncertainties.

In the context of North Macedonia, this assessment highlights three core dimensions in evaluating the applicability of AI-assisted judicial systems: ethical principles, technological infrastructure, and legal regulation. From an ethical standpoint, in order to establish public legitimacy, it is essential to build trust that AI systems are impartial and fair. In this regard, it is recommended that local ethical guidelines be developed in alignment with the Council of Europe's key principles namely, respect for human rights (Nuredin, 2022a), non-discrimination, security, transparency, and user control.

Before being implemented in courts, AI systems should undergo bias testing, and both algorithmic transparency and decision explainability must be integral to the process. As a starting point, pilot programs could be introduced in limited areas such as small claims disputes through a judge-assistant model, allowing gradual and controlled integration.

From a technological standpoint, accelerating digital transformation in North Macedonia is essential. The digitization of court records, development of integrated case management systems, facilitation of inter-institutional data sharing, and investment in cybersecurity form the foundation of this process. Citizen-oriented platforms such as an e-Court portal would not only enhance transparency but also help meet the data demands of AI systems. In this regard, EU funding and regional cooperation initiatives can provide valuable support. In terms of human resources, the recruitment of IT specialists in the public sector and the provision of continuous training should be prioritized.

On the legal front, North Macedonia must first develop a National Artificial Intelligence Strategy, followed by the adoption of a dedicated AI Law or amendments to existing legislation. The legal status of AI-assisted decisions should be clearly defined, requiring final approval by a human judge and ensuring that individuals retain access to legal remedies. Alignment with the EU's AI Act and the adaptation of best practices from countries such as Estonia, France, and the Netherlands will be critical in guiding this transition.

In conclusion, the implementation of AI judges in North Macedonia will require the strengthening of both technical and normative infrastructures. However, if this transformation is managed properly with full respect for the right to a fair trial and the principle of judicial independence it can lay the foundation for a human-centered, technology-assisted justice system. The proposed roadmap involves starting with limited pilot projects and continuously refining the system through ongoing monitoring and evaluation. In this way, AI can serve as the “assistant angel” to the human judge; but the scales of justice must always remain under the guidance of human intellect and conscience.

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